

Jacob Vanluven

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Education

Indiana University of Pennsylvania | Indiana, PA

B.S. Computer Science and Mathematics | Dean's List: Fall 2023 – Spring 2026

Expected Spring 2027

GPA: 3.55/4.00

Professional Experience

Student Worker for the Kopchick College Dean's Office

November 2023 - Present

Indiana University of Pennsylvania | Indiana, PA

- Perform ongoing website maintenance and Title II accessibility compliance audits
- Manage and update digital signage across the college using OptiSigns
- Assist students and faculty with academic needs

Technical Skills

Languages:

GDScript

C#

Python

Frameworks:

Godot 4

Unity

PyTorch

Web Development:

Next.js

Typescript

HTML/CSS

Art/Design:

Aseprite

Inkscape

Projects

DocuDINO | PyTorch, DINO, FAISS

Summer-Fall 2026

2026 U-SOAR Research Program, continued | Indiana, PA

- Developed a custom training framework specialized for writer retrieval on historical documents
- Evaluation pipeline achieved 90% top-1 accuracy on HistoricalWI dataset

Personal Website | Next.js, HTML, Tailwindv4, Typescript, Nginx

Fall 2026

Personal Project

- Self-hosted on a DigitalOcean droplet using Nginx; configured Cloudflare DNS and UFW
- Implemented a blog system and subdomain support for larger projects

Multiplayer Game | Godot, GDScript, Aseprite

Spring 2026

CMSC-473 | Software Engineering Practice (Group Project)

- Led 6 rounds of Agile sprints to complete development of a multiplayer 2D sandbox game
- Developed a streamlined tile management system, and built out an authoritative multiplayer server
- Designed and created original pixel art for tiles, characters, and over 100 unique items

Hand Recognition Control System | Python, OpenCV, MediaPipe

Summer 2025

2025 U-SOAR Research Program | Indiana, PA

- Received a competitive stipend to conduct undergraduate research with faculty advisor Dr. Grieggs
- Built a hand-tracking interface using Google's MediaPipe hand-landmark model, enabling mouse and keyboard control via vision-based gesture recognition
- Developed a client-server pipeline between Godot and the main system to explore in-game integration

Activities

IUP Competitive Programming Team

Fall 2023 - Present

- Participate in two annual competitions: the ICPC East-Central North America competition in the Fall and the PACISE competitive programming competition in the spring
- Utilize greedy algorithms and dynamic programming as the lead programmer of a 3-person Python team

Achievements

PACISE Conference

Spring 2026

Competitive Programming | Shippensburg, PA

- Achieved first place among 11 other PA state-schools